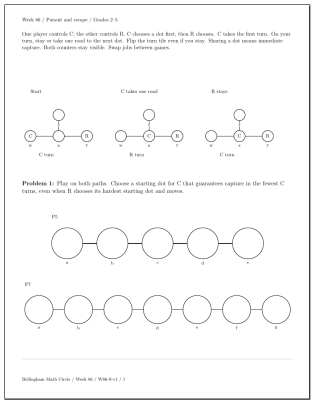
Week 86 / Pursuit and escape / Problem menu

for adults: pick, then print


Week 86 / Pursuit and escape / Grades 2–5

One player controls C, the other controls R. C chooses a dot first, then R chooses. C takes the first turn. On your turn, stay or take one road to the next dot. Flip the turn tile even if you stay. Sharing a dot means immediate capture. Both counters stay visible. Swap jobs between games.

Start C takes one road R stays

C turn R turn C turn

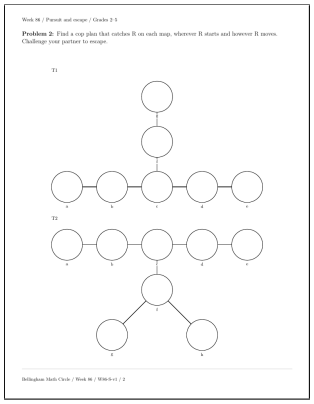
Problem 1: Play on both paths. Choose a starting dot for C that guarantees capture in the fewest C turns, even when R chooses its hardest starting dot and moves.

P5:

P7:

Bellingham Math Circle / Week 86 / W86-S-v1 / 1

Best cop start on two paths
Grades 2–5 P1 · Grades 4–5 P1
first route


Week 86 / Pursuit and escape / Grades 2–5

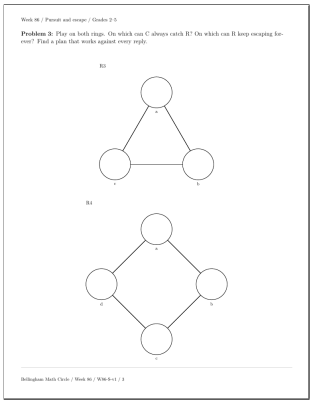
Problem 2: Find a cop plan that catches R on each map, wherever R starts and however R moves. Challenge your partner to escape.

T1

T2

Bellingham Math Circle / Week 86 / W86-S-v1 / 2

A cop plan on two trees
Grades 2–5 P2
first route


Week 86 / Pursuit and escape / Grades 2–5

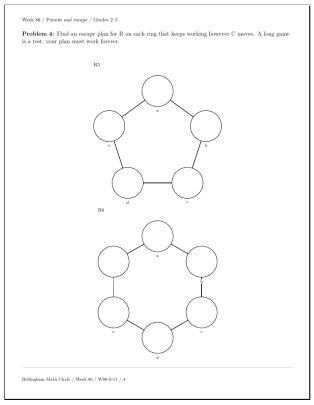
Problem 3: Play on both rings. On which can C always catch R? On which can R keep escaping forever? Find a plan that works against every reply.

R3

R4

Bellingham Math Circle / Week 86 / W86-S-v1 / 3

Triangle against square
Grades 2–5 P3
first route


Week 86 / Pursuit and escape / Grades 2–5

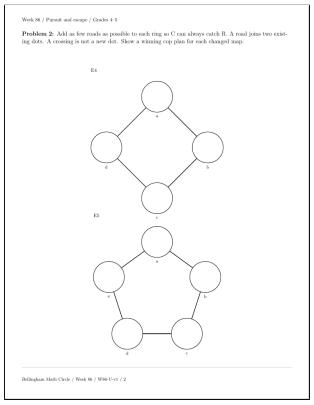
Problem 4: Find an escape plan for R on each ring that keeps working however C moves. A long game is a tool; your plan must work forever.

R5

R6

Bellingham Math Circle / Week 86 / W86-S-v1 / 4

Escape on five- and six-dot rings
Grades 2–5 P4
first route


Week 86 / Pursuit and escape / Grades 4–5

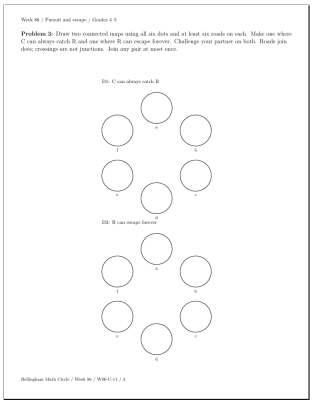
Problem 2: Add as few roads as possible to each ring so C can always catch R. A road joins two existing dots. A crossing is not a new dot. Show a winning cop plan for each changed map.

E4

E5

Bellingham Math Circle / Week 86 / W86-U-v1 / 2

Fewest added roads so C wins
Grades 4–5 P2
return route


Week 86 / Pursuit and escape / Grades 4–5

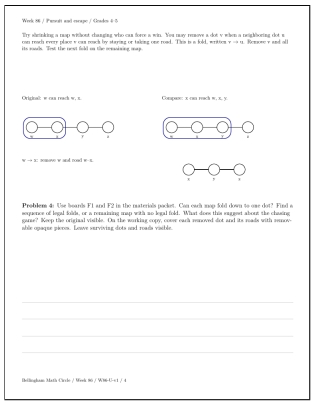
Problem 3: Draw two connected maps using all six dots and at least six roads on each. Make one where C can always catch R and one where R can escape forever. Challenge your partner on both. Roads join dots; crossings are not junctions. Join any pair at most once.

D3: C can always catch R

D3: R can escape forever

Bellingham Math Circle / Week 86 / W86-U-v1 / 3

Design a cop-win and a robber-win map
Grades 4–5 P3
return route


Week 86 / Pursuit and escape / Grades 4–5

Try shrinking a map without changing who can force a win. You may remove a dot v when a neighboring dot u can reach every place v can reach by staying or taking one road. This is a fold, written $v \to u$. Remove v and all its roads. Test the next fold on the remaining map.

Original: w can reach w, x.

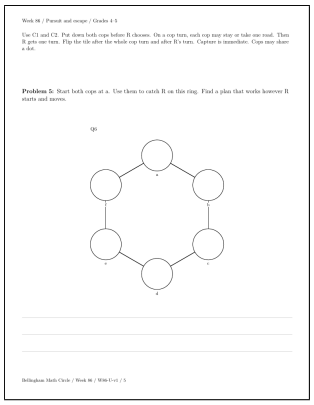
Compare: x can reach w, x, y.

$w \to x$: remove w and road w–x.

Problem 4: Use boards F1 and F2 in the materials packet. Can each map fold down to one dot? Find a sequence of legal folds, or a remaining map with no legal fold. What does this suggest about the chasing game? Keep the original visible. On the working copy, cover each removed dot and its roads with removable opaque pieces. Leave surviving dots and roads visible.

Bellingham Math Circle / Week 86 / W86-U-v1 / 4

Fold F1 and F2 down
Grades 4–5 P4
return route


Week 86 / Pursuit and escape / Grades 4–5

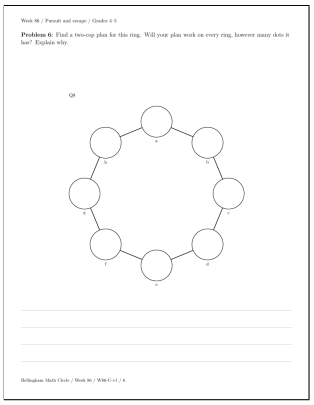
Use C1 and C2. Put down both cops before R chooses. On a cop turn, each cop may stay or take one road. Then R gets one turn. Flip the tile after the whole cop turn and after R's turn. Capture is immediate. Cops may share a dot.

Problem 5: Start both cops at a. Use them to catch R on this ring. Find a plan that works however R starts and moves.

Q5

Bellingham Math Circle / Week 86 / W86-U-v1 / 5

Two cops from a on the six-ring
Grades 4–5 P5
return route


Week 86 / Pursuit and escape / Grades 4–5

Problem 6: Find a two-cop plan for this ring. Will your plan work on every ring, however many dots it has? Explain why.

Q6

Bellingham Math Circle / Week 86 / W86-U-v1 / 6

Two cops on the eight-ring and on every ring
Grades 4–5 P6
return route